

ENG-654 · LECTURE 07

Workspace Path Planning

Follow a supplied workspace path or trajectory with a feasible joint motion

The desired motion is supplied by a precomputed trajectory or a real-time policy. We study planning methods for noncuspidal and cuspidal robots.

We study workspace path planning with a given path or trajectory

Assume that a desired workspace path or trajectory has been given to the robot to follow. It can come from either source below.

1 · Precomputed trajectories

The desired motion is calculated in advance and supplied as a sequence of workspace targets, with timing when a trajectory is specified.

The complete command is available for analysis before execution.

2 · Real-time policies

A policy uses current observations to provide the next desired workspace pose or motion command while the robot moves.

Only the current command or a short future horizon may be available.

A path specifies where to move; a trajectory also specifies when. Our task is to find a continuous joint motion that follows the supplied command while checking singularities and joint limits.

Find feasible joint motions for the supplied workspace command

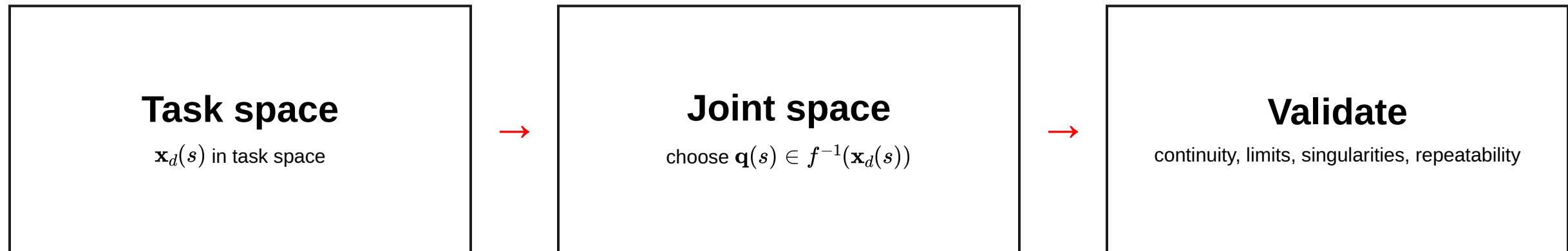
01 **Numerical and analytical planning**
Local continuation, complete inverse solutions, and connectivity.

02 **Noncuspidal robots · ABB IRB 4600**
Hybrid planning or analytical shoulder–elbow–wrist classification.

03 **Cuspidal 3R robots · custom_3R.urdf**
One task-space curve can produce different joint-space outcomes.

04 **Cuspidal 6R robots · ABB GoFa CRB 15000**
Inverse-solution maps, joint limits, and connections between configurations.

Relation between task space and joint space



Forward kinematics maps joint configurations to task poses: $f : \mathcal{Q} \rightarrow \mathcal{X}$. A **lift** is a continuous joint path with $f(\mathbf{q}(s)) = \mathbf{x}_d(s)$. The **same loop** means the identical closed task-space curve, $\mathbf{x}_d(0) = \mathbf{x}_d(1)$; different starting IKs can give different joint paths, and $\mathbf{q}(0) = \mathbf{q}(1)$ need not hold.

Build a joint path using analytical IK or numerical continuation

Analytical or algebraic IK

$$\mathcal{Q}_k = f^{-1}({}^0T_{t,d}(t_k))$$

Enumerate every real solution at each target pose using closed-form expressions or an algebraic root solver. Connect adjacent solutions continuously.

Enumeration reveals candidates; connectivity and constraints determine feasible paths.

Numerical continuation

$${}^0\mathcal{V}_t = \begin{bmatrix} {}^0\dot{\mathbf{p}}_t \\ {}^0\boldsymbol{\omega}_t \end{bmatrix} = {}^0J_t(\mathbf{q})\dot{\mathbf{q}}$$

Solve the velocity equation near the current configuration. Predict the next joint state, correct its FK error, and validate that step.

One initial IK selects one local trajectory.

For a 6R robot, “ $\dot{\mathbf{x}} = J\dot{\mathbf{q}}$ ” uses the geometric velocity $[\dot{\mathbf{p}}; \boldsymbol{\omega}]$. An Euler-angle derivative needs its representation-dependent angular mapping.

Enumeration gives global candidates; continuation gives local tracking

	Jacobian / numerical	Analytical branch evaluation
Information used	Local derivative at the current $\mathbf{q}$	Closed-form or algebraic inverse structure for the modeled geometry
Initialization	A seed configuration	Desired pose and robot parameters; labels organize the returned roots
Output	One nearby solution, if convergence succeeds	All evaluated real roots, then a selected branch
Singularity signal	Conditioning degrades during iteration	Branch discriminant or separator vanishes
Scope	LOCAL TRACKING	GLOBAL ROOT ENUMERATION, NOT GLOBAL PATH CERTIFICATION

Local safeguards detect failure; they cannot select a globally viable start

Step size

Large Cartesian steps invalidate the local linear model and can jump to another basin.

Near singularity

Stop when the full Jacobian becomes singular or ill-conditioned. Damping can moderate a numerical step, but cannot restore a missing task direction.

Joint limits

A regular branch may still hit a physical joint stop. Global comparison can select a different initial IK that completes the path; it does not remove the stop.

$\|\dot{\mathbf{q}}\|$ can grow while $\|\dot{\mathbf{x}}\|$ stays modest as $\sigma_{\min}(\mathbf{J}) \rightarrow 0$.

ABB IRB 4600 · IK solutions

ABB IRB 4600 · starting tool pose

Start: (1.40, 0.30, 1.10) m · fixed downward tool orientation.
Eight mathematical IK solutions; four inside joint limits. Select an IK to view it.

Joint angles in degrees · determinant in m³

IK	S/E/W	q1	q2	q3	q4	q5	q6	det J	Limits
1	S+ E+ W+	12.1	1.3	22.6	-0.0	66.1	-167.9	-1.777	Within
2	S+ E+ W-	12.1	1.3	22.6	180.0	-66.1	12.1	1.777	Within
3	S+ E- W+	12.1	117.7	173.1	-0.0	159.2	-167.9	0.690	Outside
4	S+ E- W-	12.1	117.7	173.1	180.0	-159.2	12.1	-0.690	Outside
5	S- E+ W+	-167.9	-111.4	2.0	180.0	160.6	-167.9	0.665	Outside
6	S- E+ W-	-167.9	-111.4	2.0	-0.0	-160.6	12.1	-0.665	Outside
7	S- E- W+	-167.9	-19.1	-166.3	180.0	84.6	-167.9	-1.990	Within
8	S- E- W-	-167.9	-19.1	-166.3	-0.0	-84.6	12.1	1.990	Within

Recall: $\det {}^0J_6 = -a_2FG \sin q_5$, $S = \operatorname{sgn} G$, $E = \operatorname{sgn} F$, $W = \operatorname{sgn}(\sin q_5)$.
 $F = a_3 \sin q_3 + d_4 \cos q_3$, $G = a_1 + a_2 \sin q_2 + a_3 \sin(q_2 + q_3) + d_4 \cos(q_2 + q_3)$.
 $G = 0$: shoulder · $F = 0$: elbow · $\sin q_5 = 0$: wrist singularity.

Start with all analytical IKs, then follow each with the Jacobian

Initialize: solve $\mathcal{Q}_0 = f^{-1}(\mathbf{x}_d(t_0))$ analytically; classify S/E/W and keep starts within joint limits.

for each initial configuration $\mathbf{q}_0 \in \mathcal{Q}_0$:

for $k = 0, \dots, N - 1$:

Predict: solve ${}^0J_6\dot{\mathbf{q}}_k = {}^0\mathcal{V}_{6,d}$; set $\hat{\mathbf{q}}_{k+1} = \mathbf{q}_k + \Delta t\dot{\mathbf{q}}_k$.

Correct: iterate Newton updates at the next target pose until the FK error is small.

Validate: stop this branch on failed correction, singularity, or joint-limit crossing. Check speed demands; increase duration if needed.

end for · store this branch's outcome.

end for · compare complete paths and choose a feasible start.

$$\begin{bmatrix} {}^0\mathbf{v}_6 \\ {}^0\boldsymbol{\omega}_6 \end{bmatrix} = {}^0J_6(\mathbf{q})\dot{\mathbf{q}}$$

$${}^0J_6(\mathbf{q}_k)\Delta\mathbf{q} = \begin{bmatrix} {}^0\mathbf{p}_{6,d} - {}^0\mathbf{p}_6 \\ \text{Log}(R_d R^\top)^\vee \end{bmatrix}$$

$$|\det {}^0J_6| < \varepsilon \implies \text{stop this continuation}$$

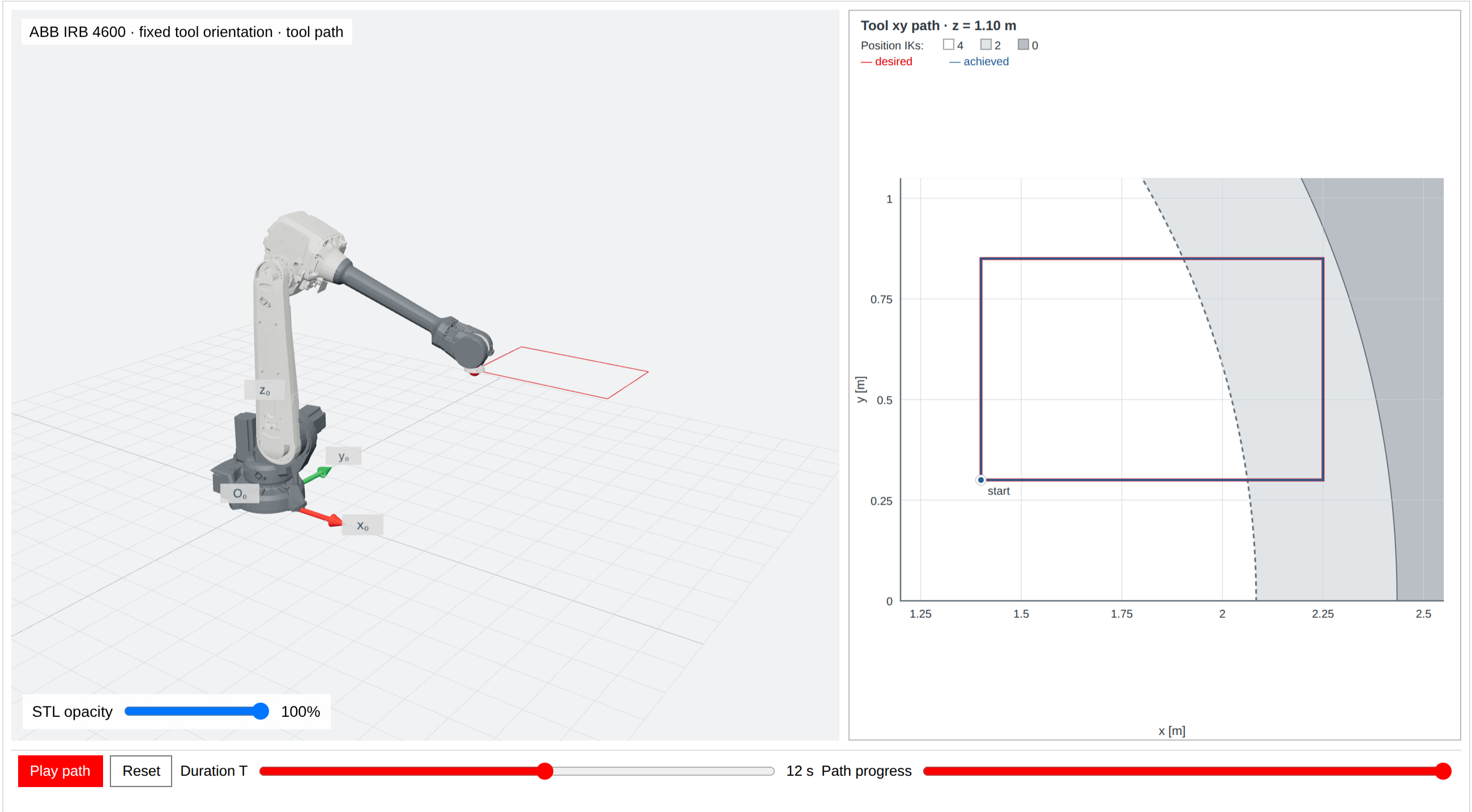
Evaluate the twist at the tool. For this robot, the tool origin is O_6 . With ${}^0\mathbf{r}_{56} = {}^0\mathbf{p}_6 - {}^0\mathbf{p}_5$, shift the wrist

Jacobian to the tool: ${}^0J_6 = \begin{bmatrix} I & -[{}^0\mathbf{r}_{56}]_\times \\ 0 & I \end{bmatrix} {}^0J_5$. Here

$[\mathbf{r}]_\times \mathbf{a} = \mathbf{r} \times \mathbf{a}$, and $\det {}^0J_6 = \det {}^0J_5$.

Here $R = {}^0R_{\text{tool}}$ and $R_d = {}^0R_{\text{tool},d}$ are the current and desired tool orientations. $\text{Log}(R_d R^\top)^\vee$ is their axis-angle error vector in the world frame.

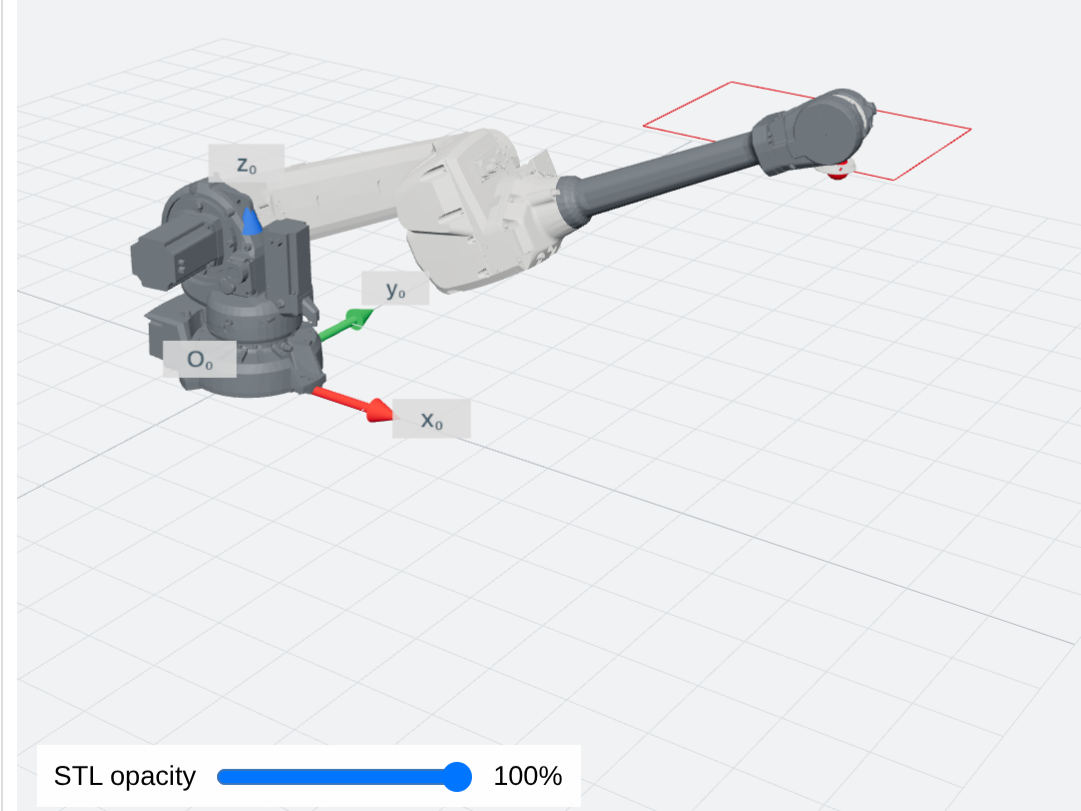
An xy rectangle crosses regions with different numbers of positional IKs



Four legal starts: two finish; two reach a positional singularity

Analytical IK supplies the starting solutions; each numerical predictor and correction uses the current Jacobian. Compare every starting IK on the same rectangle: successful local convergence at the start does not ensure completion.

ABB IRB 4600 · fixed tool orientation · tool path

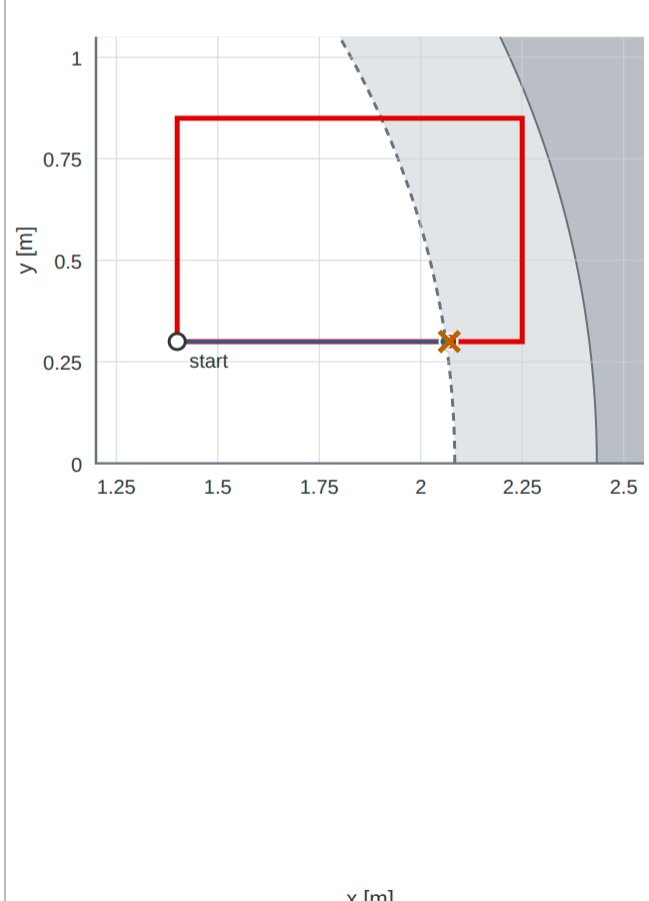


STL opacity 100%

Tool xy path · z = 1.10 m

Position IKs: 4 2 0

— desired — achieved



x [m]

Play path

Reset

S- E- W+

Duration T 12 s

Path progress

Start IK	Local continuation
S+ E+ W+	Tracking
S+ E+ W-	Tracking
S+ E- W+	Rejected: q3 > 75°
S+ E- W-	Rejected: q3 > 75°
S- E+ W+	Rejected: q2 < -90°
S- E+ W-	Rejected: q2 < -90°
S- E- W+	Stopped: The selected shoulder/elbow branch ends at the 4 → 2 positional-IK boundary
S- E- W-	Stopped: The selected shoulder/elbow branch ends at the 4 → 2 positional-IK boundary

s = 0.197 · S- E- W+ · hybrid predictor + Newton

2 position IKs · 4 full-pose IKs at the desired point

q1...q6 [°]: -171.7, -70.1, -85.5, 180.0, 114.5, -171.7

|det °J6| = 0.1545 m³ · FK error 2.8e-12

q̇ [°/s]: -1.7, -367.4, 675.2, -0.0, 307.8, -1.7

T=12s · peak speed / limit = 3.86 – increase T (sampled admissible prefix)

This branch reaches a positional fold. The selected shoulder/elbow branch ends at the 4 → 2 positional-IK boundary. The robot holds its last admissible configuration. Choose S+ E+ to complete the same command.

Compare all ABB solution branches along the entire rectangle

Analytical IK enumerates the full-pose roots at each path sample. Continuous branch tracking and joint-limit checks identify the initial configurations that can finish.

ABB IRB 4600 · fixed tool orientation · tool path

STL opacity 100%

Tool xy path · z = 1.10 m

Position IKs: 4 2 0
— desired — achieved

x [m]

Play path

Reset

S+ E+ W+ ▼

Duration T 12 s

Path progress

Start IK	Whole-path analysis
S+ E+ W+	Completes loop
S+ E+ W-	Completes loop
S+ E- W+	Rejected: q3 > 75°
S+ E- W-	Rejected: q3 > 75°
S- E+ W+	Rejected: q2 < -90°
S- E+ W-	Rejected: q2 < -90°
S- E- W+	Stops s=0.197: The selected shoulder/elbow branch ends at the 4 → 2 positional-IK boundary
S- E- W-	Stops s=0.197: The selected shoulder/elbow branch ends at the 4 → 2 positional-IK boundary

s = 1.000 · S+ E+ W+ · analytical IK

4 position IKs · 8 full-pose IKs at the desired point

q1...q6 [°]: 12.1, 1.3, 22.6, -0.0, 66.1, -167.9

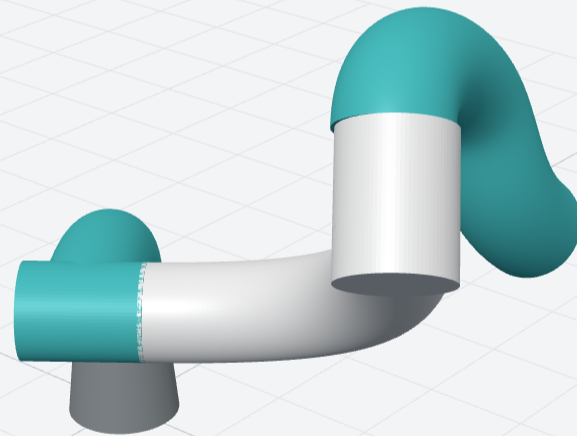
|det °J6| = 1.7768 m³ · FK error 3.6e-16

q̇ [°/s]: 0.0, 0.0, 0.0, 0.0, 0.0, 0.0

T=12s · peak speed / limit = 0.26 ✓ (sampled whole path)

8 mathematical starts → 4 admissible starts → 2 complete sampled joint paths. All IK roots are recomputed at every pose; limits and continuation decide which starts work.

We are going to study cuspidal robots



STL opacity 100%

Edit one path; carry its exact vertices into the following slides

One shared task-space path · $y = 0$

-

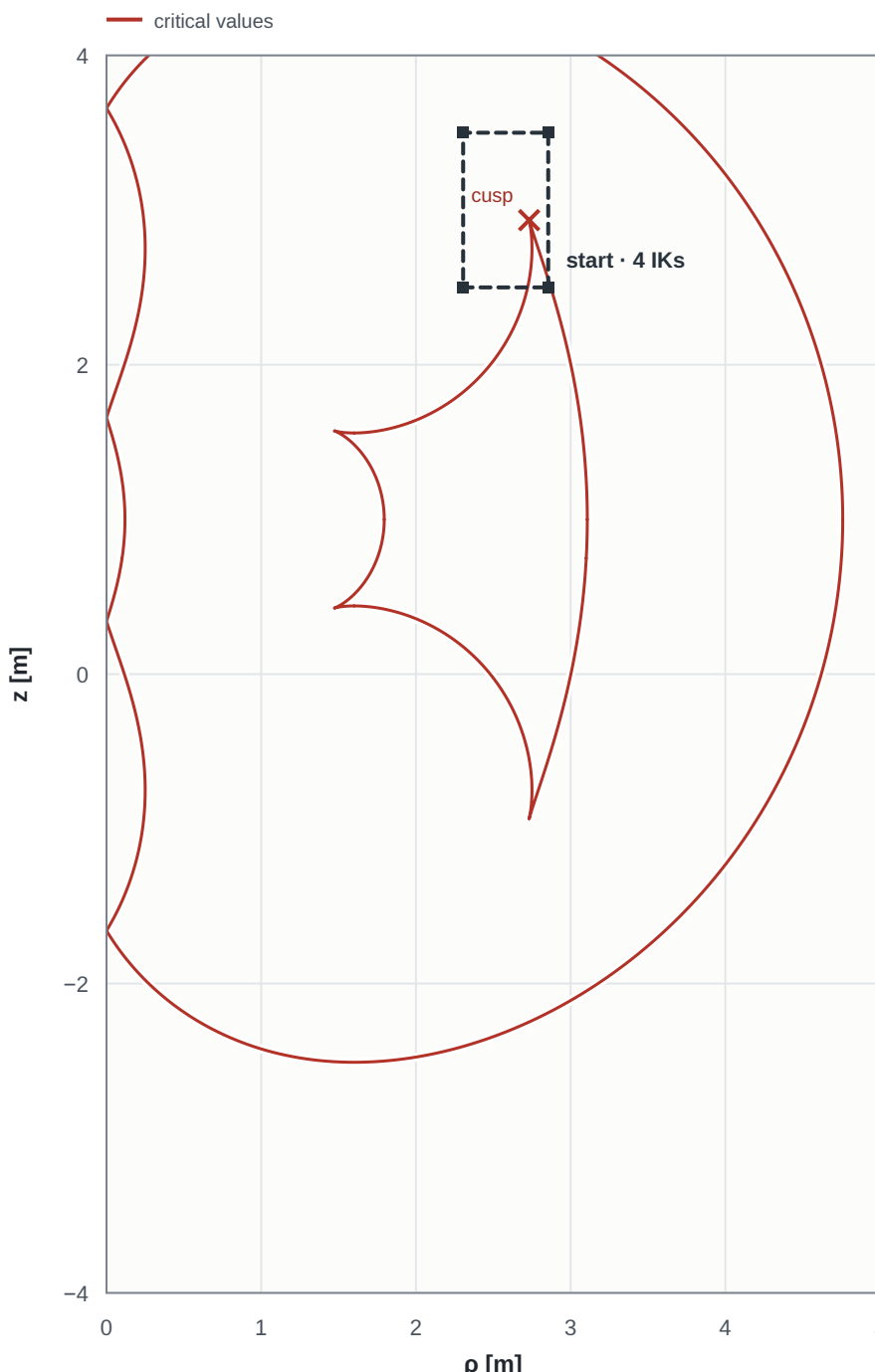
+

Reset axes

Fit path / cusp

Wheel: zoom · Shift-drag: pan

critical values



Finalize once: every following comparison uses this exact path.

center ρ

2.58 m

center z

3.00 m

width

1.00 m

height

0.55 m

rotation

90 °

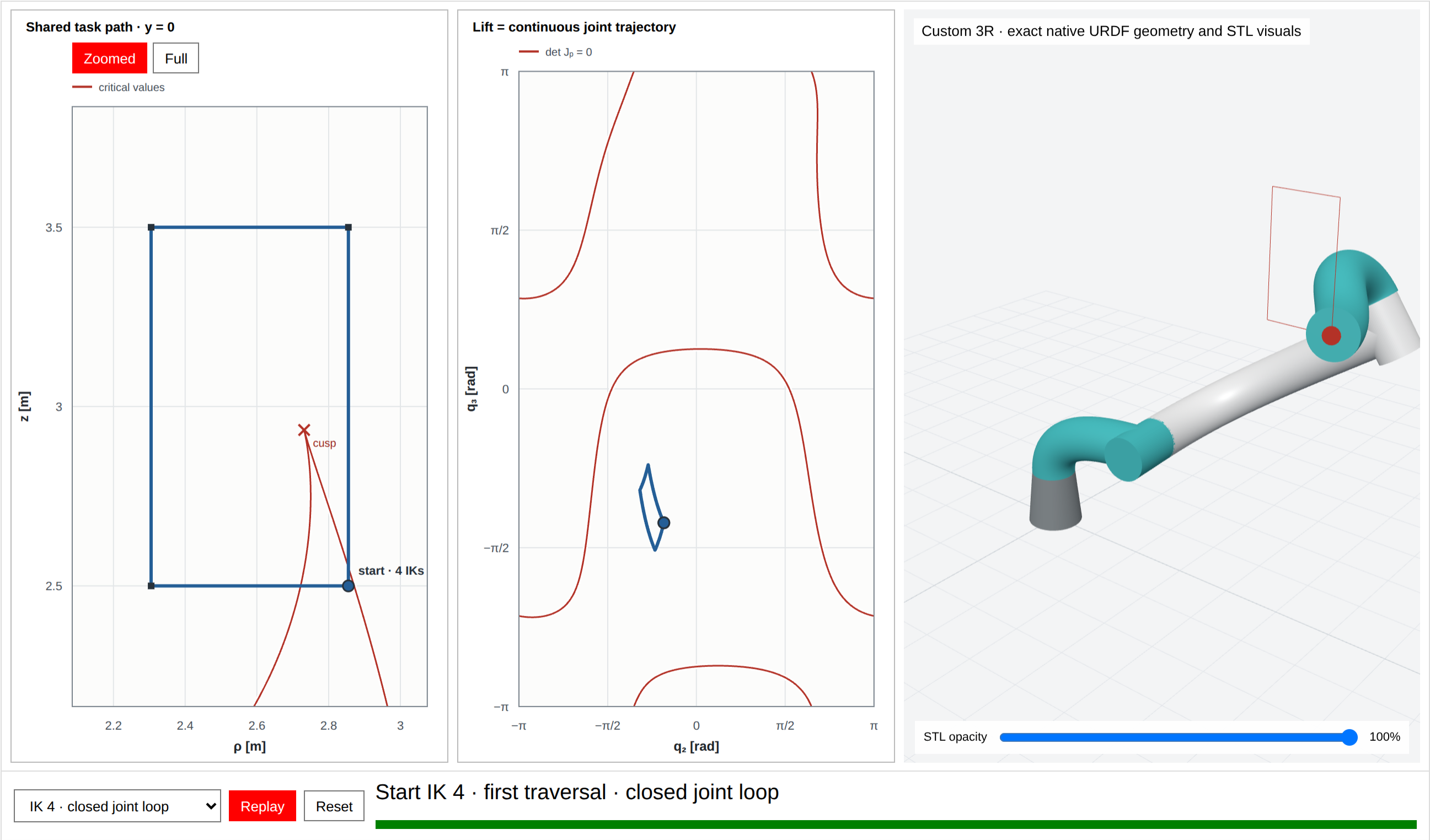
Draw vertices

Undo vertex

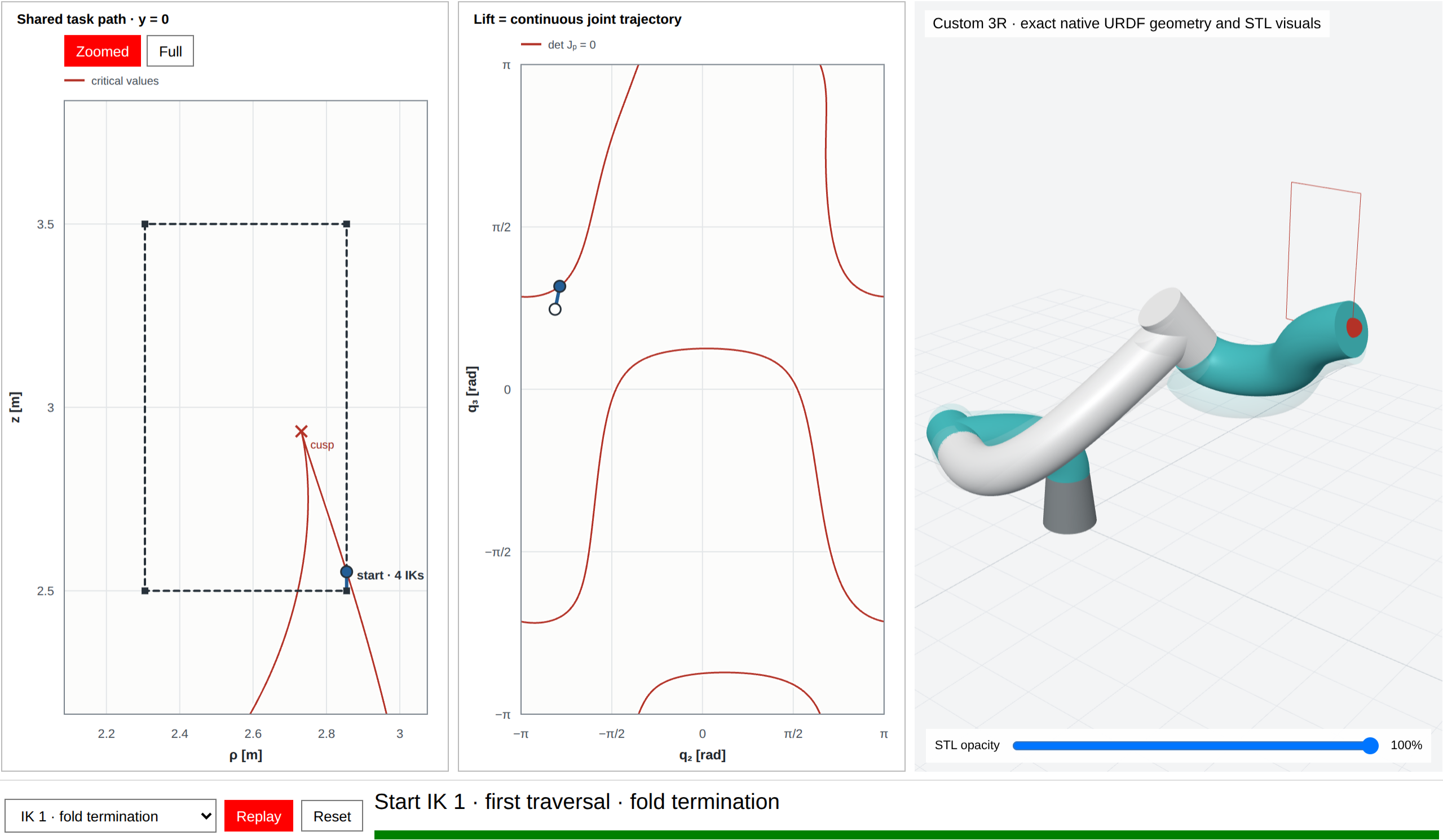
Finalize shared path

Verified example

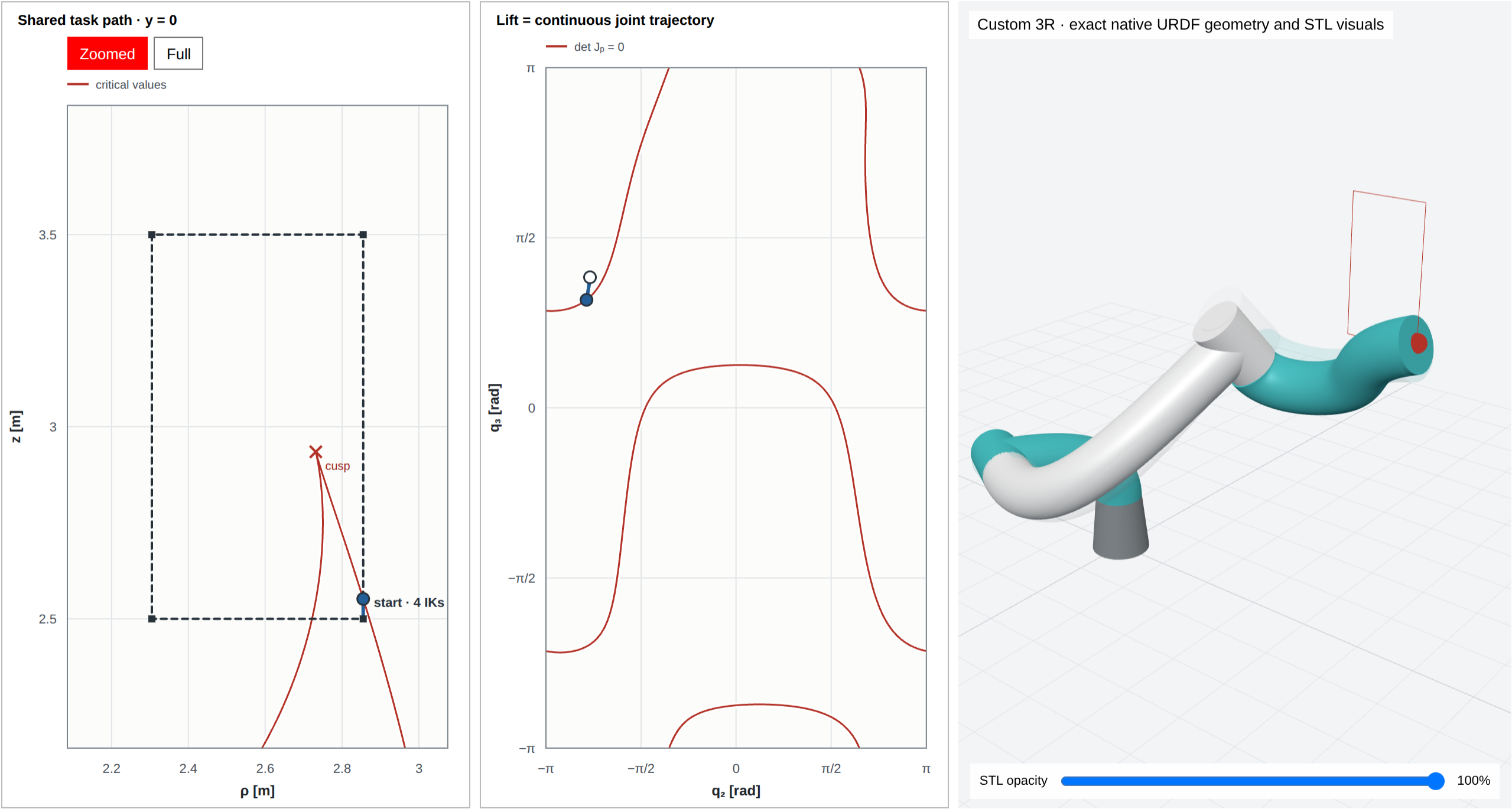
A closed task-space loop can return to the same joint configuration



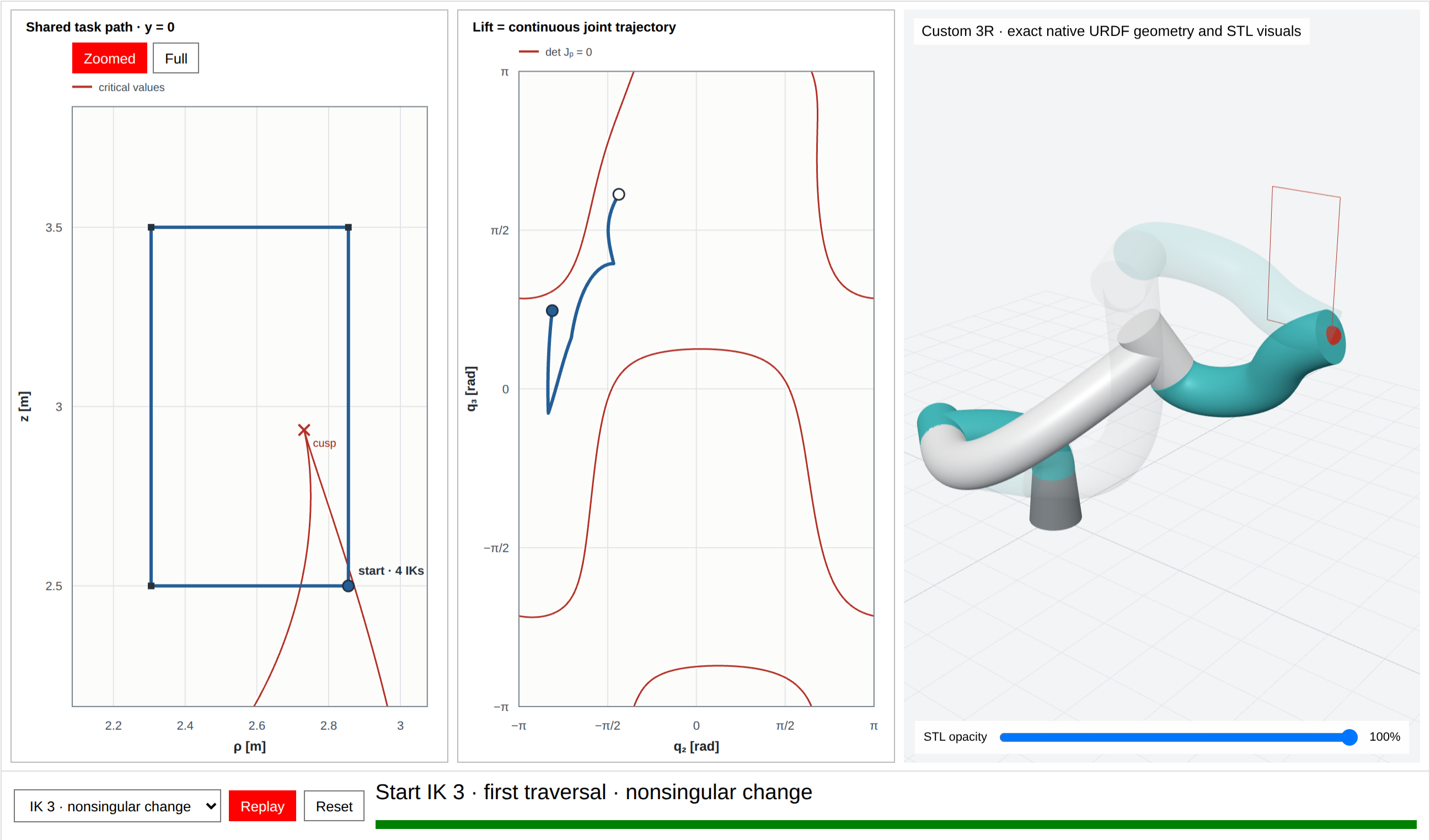
Track a starting IK whose continuation cannot complete the path



Compare a second starting IK on exactly the same task-space path



One traversal changes IK solution without crossing a singularity



IK 3 · nonsingular change

Replay

Reset

Start IK 3 · first traversal · nonsingular change

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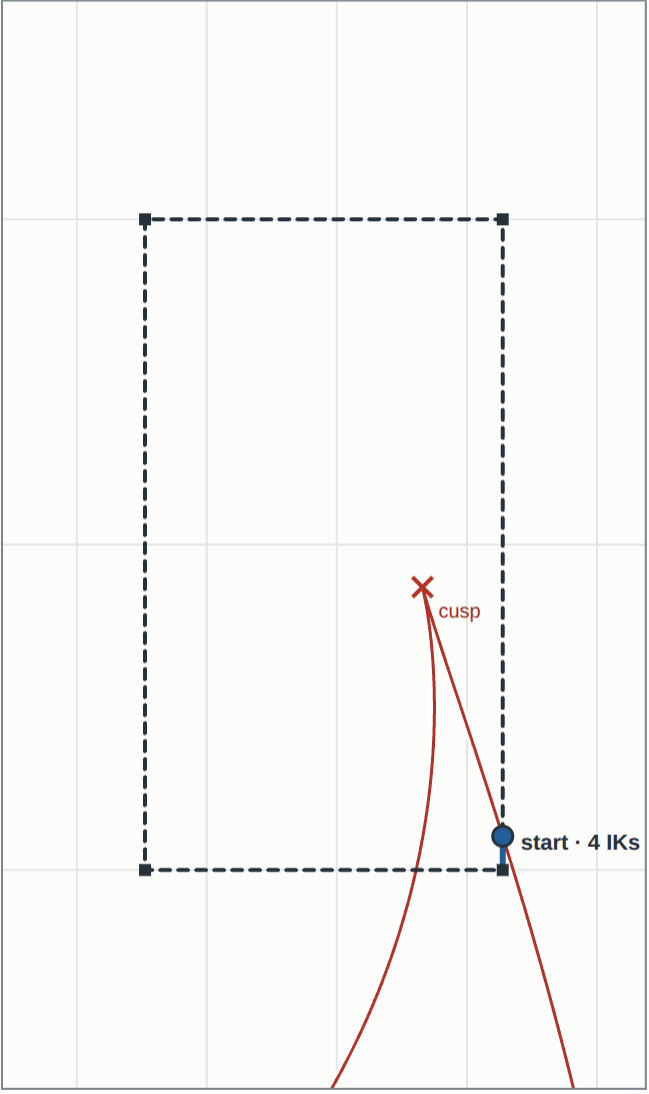
The second lap starts from the configuration reached after lap one

Shared task path · $y = 0$

Zoomed

Full

critical values



z [m]

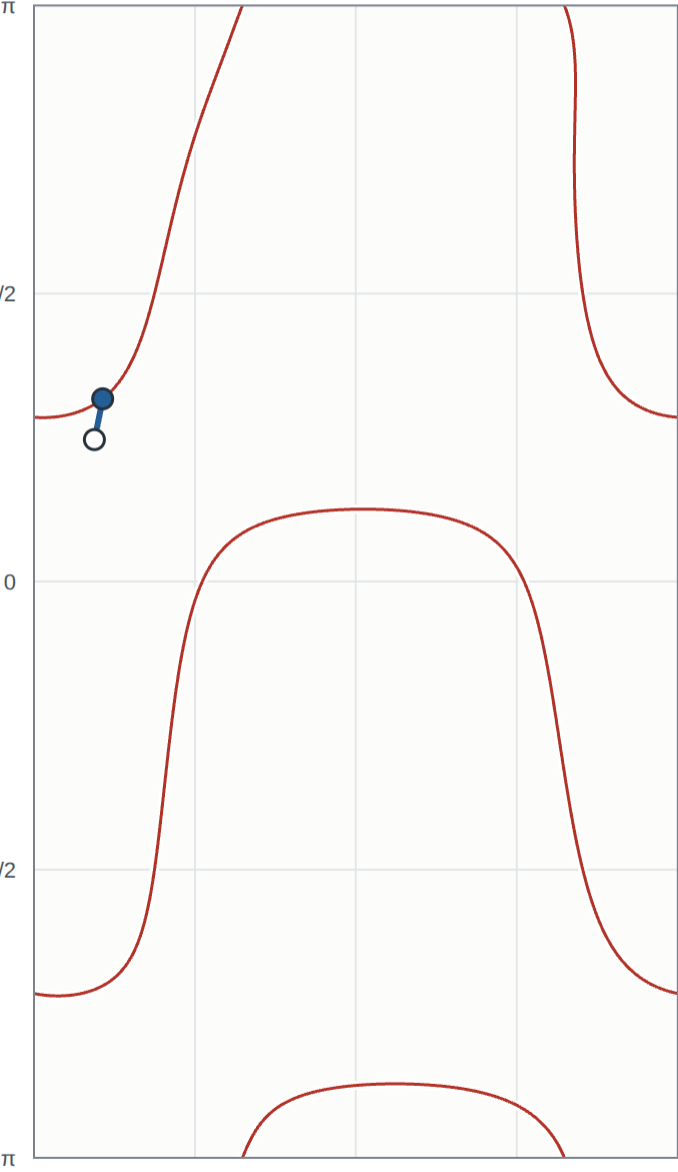
ρ [m]

start · 4 IKs

cusp

Lift = continuous joint trajectory

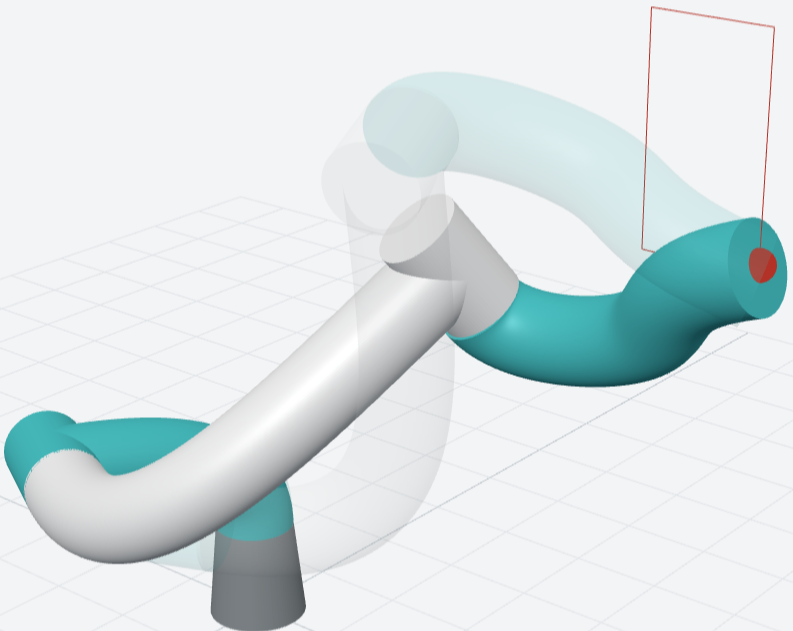
$\det J_p = 0$



q_3 [rad]

q_2 [rad]

Custom 3R · exact native URDF geometry and STL visuals



STL opacity

100%

IK 3 · nonsingular change

Replay

Reset

Start IK 3 · second traversal · fold termination

We study three types of paths in cuspidal robots

Regular

The selected joint path remains nonsingular and closes.



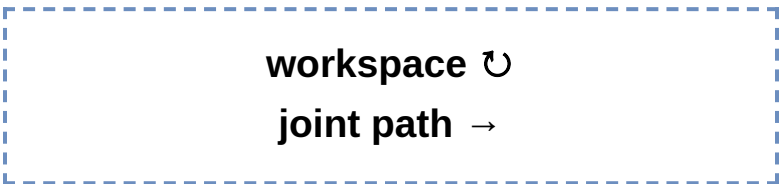
Branch-infeasible

The selected joint trajectory reaches a fold or another continuation barrier; a different starting IK may remain feasible.



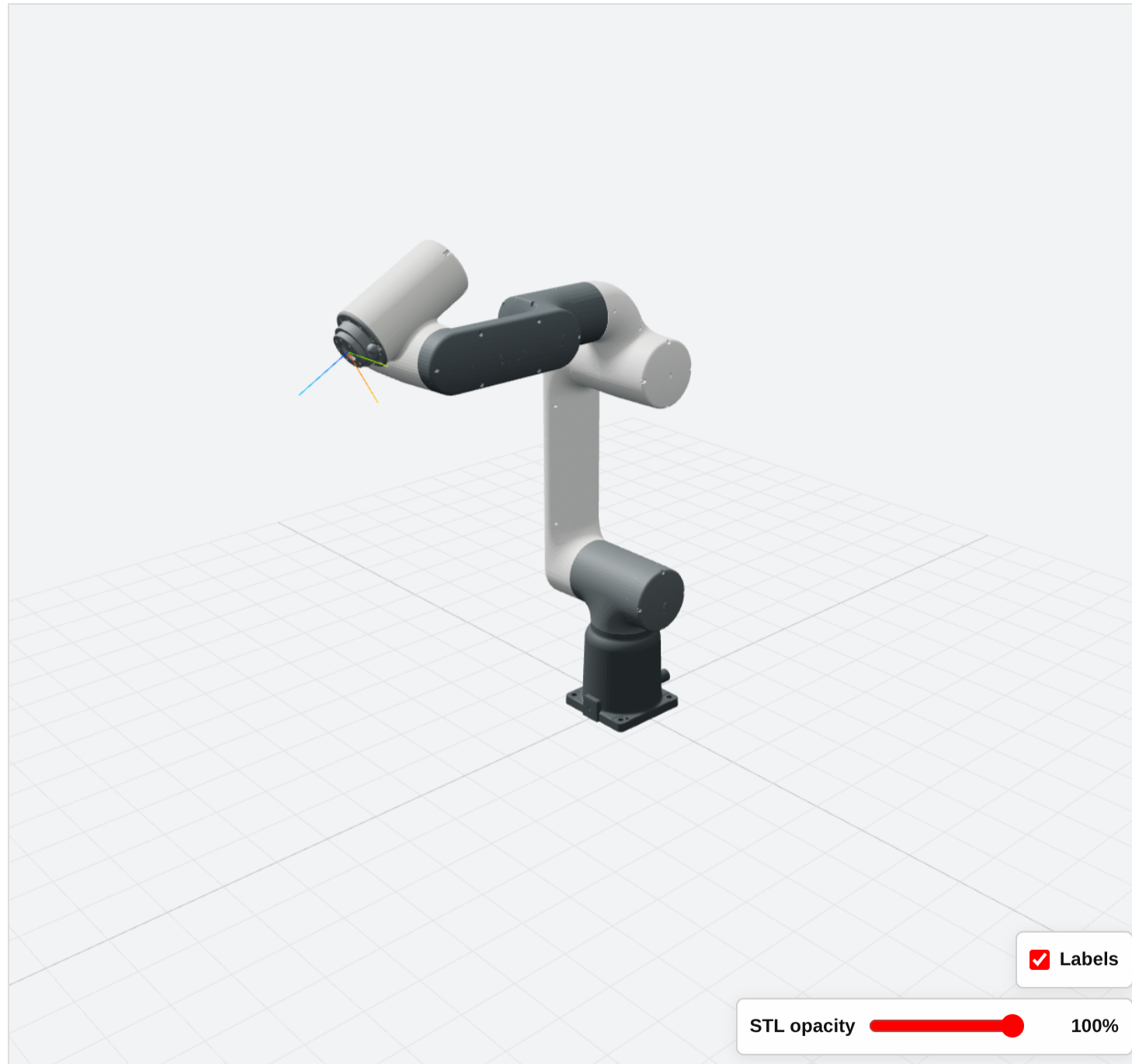
Nonsingular change

The joint path stays regular but returns to a different IK of the starting point.



Four starting IKs of the same task-space loop illustrate three types: one regular closed path, two branch-infeasible paths, and one nonsingular change of solution.

A wrist offset can change IK model



Six axes, offset wrist

The supplied CRB 15000-5/0.95 has a 0.080 m offset between its last wrist axes. The three wrist axes do not meet at one common centre.

Enumerate the configurations

The uploaded analytical IK evaluates a degree-16 polynomial, reconstructs the joint angles, and verifies each result by forward kinematics. We count real solutions and joint-limit-admissible solutions separately.

Track connectivity

Numerical labels such as “IK 1” are local to one pose. Follow the continuous joint trajectory to discover which solution it reaches at the end.

The joint-limit interval matters: $q_3 \in [-225^\circ, 85^\circ]$. A legal configuration below -180° must not be discarded merely because an equivalent angle was wrapped into another interval.

Count the inverse solutions in an x–y slice

The default map samples 100,000 poses at $z = 0.50$ m with fixed orientation $R_z(0)R_y(-0.28)R_x(1.47)$ (radians). It contains regions with 4, 6, 8, 10, 12, 14 and 16 real IKs before joint limits are applied.

Explore both scales: Central detail loads a saved 100,000-pose map for the default slice to resolve the 14- and 16-IK regions. Other heights and orientations are calculated on demand. Click a point to enumerate its solutions; compare the counts before and after joint limits.

Tool position · x-y slice · z = 0.5 m

Full slice

Central detail

☒ q₄ limit boundary

Dashed red: q₄ = ±180° for at least one legal IK. Other IKs can remain feasible across this boundary.

ABB CRB 15000 · white/grey CAD · selected IK solid

STL opacity 100%

z [m]

0.5

Yaw [°]

0

Pitch [°]

-16.042818

Roll [°]

84.224796

Map

Detailed · 100,000 poses

Calculate slice

Verified default slice

Count

All real IKs

Show all IKs

Selected only

8 IKs

10 IKs

12 IKs

14 IKs

16 IKs

Select point

Draw path

Example loop

Close loop

Undo vertex

Clear path

Track selected IK

Compare all starts

Central detail loaded · 100,000 saved poses. 16 verified IKs · 10 within joint limits.

► Joint limits and branch continuity

► Orientation and sampling

☒ IK 1

☒ IK 2

☒ IK 3 · outside limits

☒ IK 4

☒ IK 5 · outside limits

☒ IK 6

☒ IK 7

☒ IK 8

☒ IK 9

☒ IK 10 · outside limits

☒ IK 11

☒ IK 12 · outside limits

☒ IK 13

☒ IK 14 · outside limits

☒ IK 15 · outside limits

☒ IK 16

Play

Reset

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Draw one task path; compare the joint trajectories from its IKs

Choose the tool orientation and height z , then calculate the x–y map. The selected point and drawn path carry forward from the previous map. Each path holds the chosen orientation and height constant.

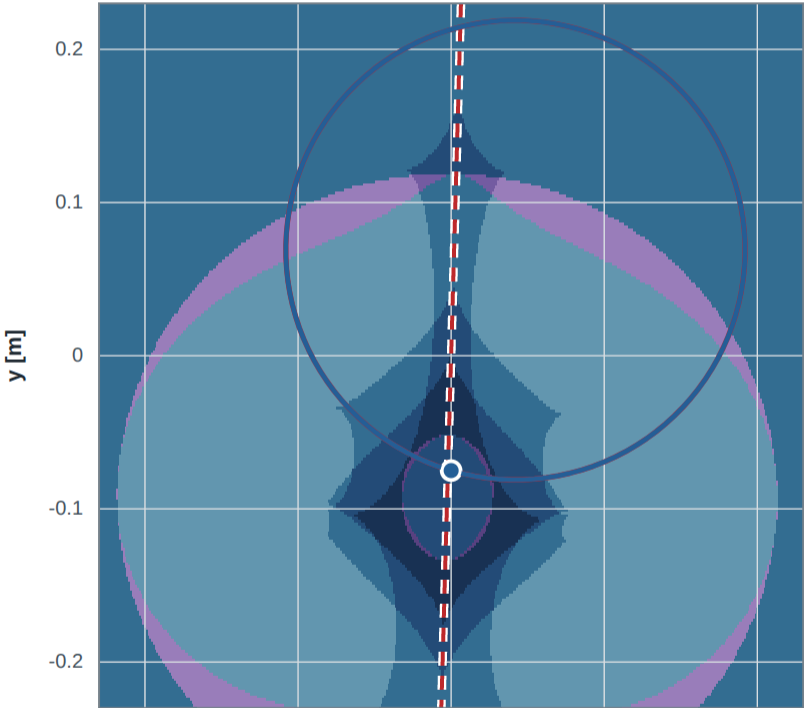
Compare the starts: at the default height and orientation, the 30 cm example loop crosses a region with eight IKs inside joint limits. Follow each starting IK: the map count alone cannot certify a continuous legal path. Inspect why each continuation stops.

Tool position · x-y slice · $z = 0.5$ m

Full slice

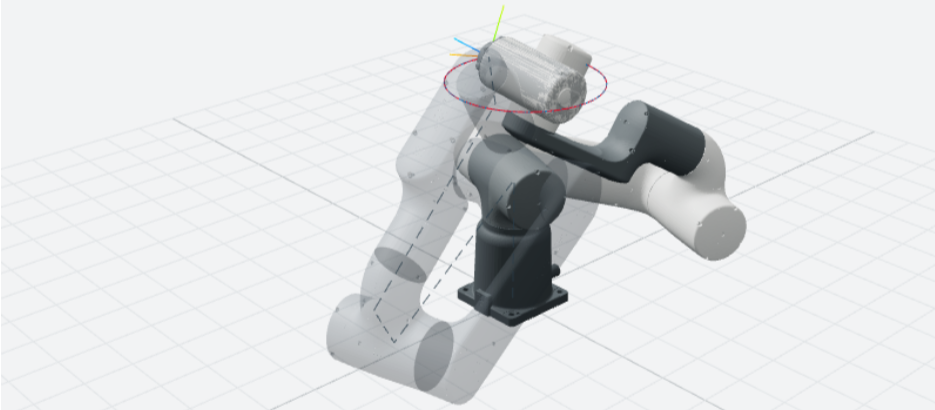
Central detail

☒ q_4 limit boundary



Dashed red: $q_4 = \pm 180^\circ$ for at least one legal IK. Other IKs can remain feasible across this boundary.

ABB CRB 15000 · white/grey CAD · selected IK solid



☒ Labels

STL opacity 100%

z [m]

0.5

Yaw [°]

0

Pitch [°]

-16.042818

Roll [°]

84.224796

Map

Detailed · 100,000 poses

Calculate slice

Verified default slice

Count

Inside joint limits

Show all IKs

Selected only

6 IKs

7 IKs

8 IKs

9 IKs

10 IKs

11 IKs

12 IKs

Select point

Draw path

Example loop

Close loop

Undo vertex

Clear path

Track selected IK

Compare all starts

Central detail loaded · 100,000 saved poses. 3/16 starting IKs complete the path. Select a result to animate it. Selected IK 1: The selected branch follows the full path within the tested joint-limit and nonsingularity margins.

► Joint limits and branch continuity

► Orientation and sampling

IK 1: complete

☒ IK 1

☒ IK 2

☒ IK 3 · outside limits

☒ IK 4

☒ IK 5 · outside limits

☒ IK 6

☒ IK 7

☒ IK 8

☒ IK 9

☒ IK 10 · outside limits

☒ IK 11

☒ IK 12 · outside limits

☒ IK 13

☒ IK 14 · outside limits

☒ IK 15 · outside limits

☒ IK 16

Replay

Reset

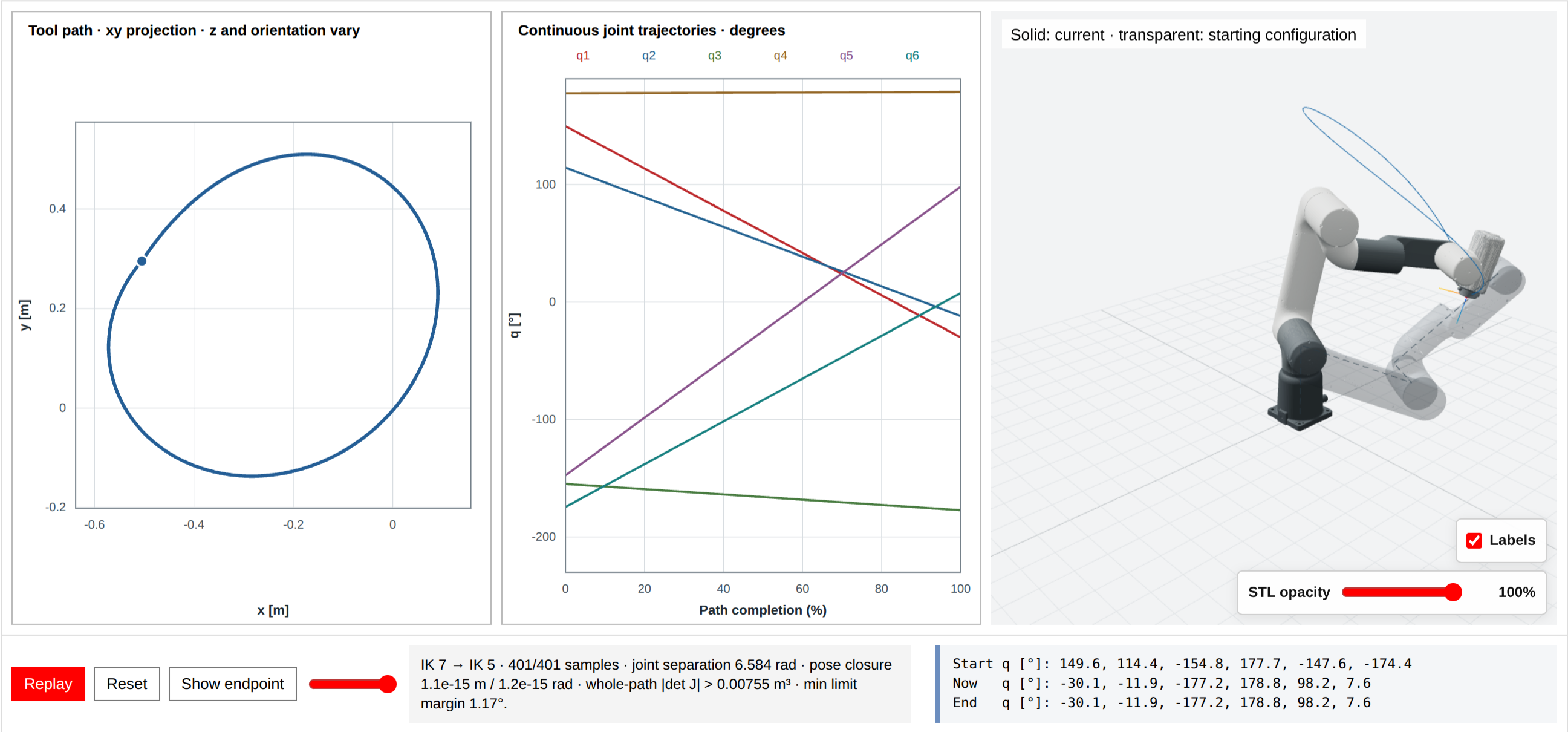
361/361 · s = 1.000

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Same final tool pose, a different joint configuration, no joint-limit crossing

A searched connection joins two IKs of the same full pose. This demonstration allows orientation to vary along the task-space loop. The transparent robot stays at the starting configuration.

Search: enumerate IK pairs of the same pose, connect legal joint configurations, and map the connection through FK. Here a determinant derivative bound verifies the entire joint segment; all 401 animation poses are rechecked by IK.



Global analysis connects solutions before choosing the joint trajectory

Receive the workspace path or trajectory from a precomputed command or a real-time policy.

Enumerate candidate IKs and connect continuous joint trajectories over the available path or planning horizon.

Track branch identity, singularity distance, joint limits, and collisions.

For closed task loops, test joint-space closure and repeatability explicitly.

Use analytical planning whenever a global property is to be studied.

Enumerate the IK solutions and analyze their continuous connections to assess path feasibility, solution changes, and repeatability.